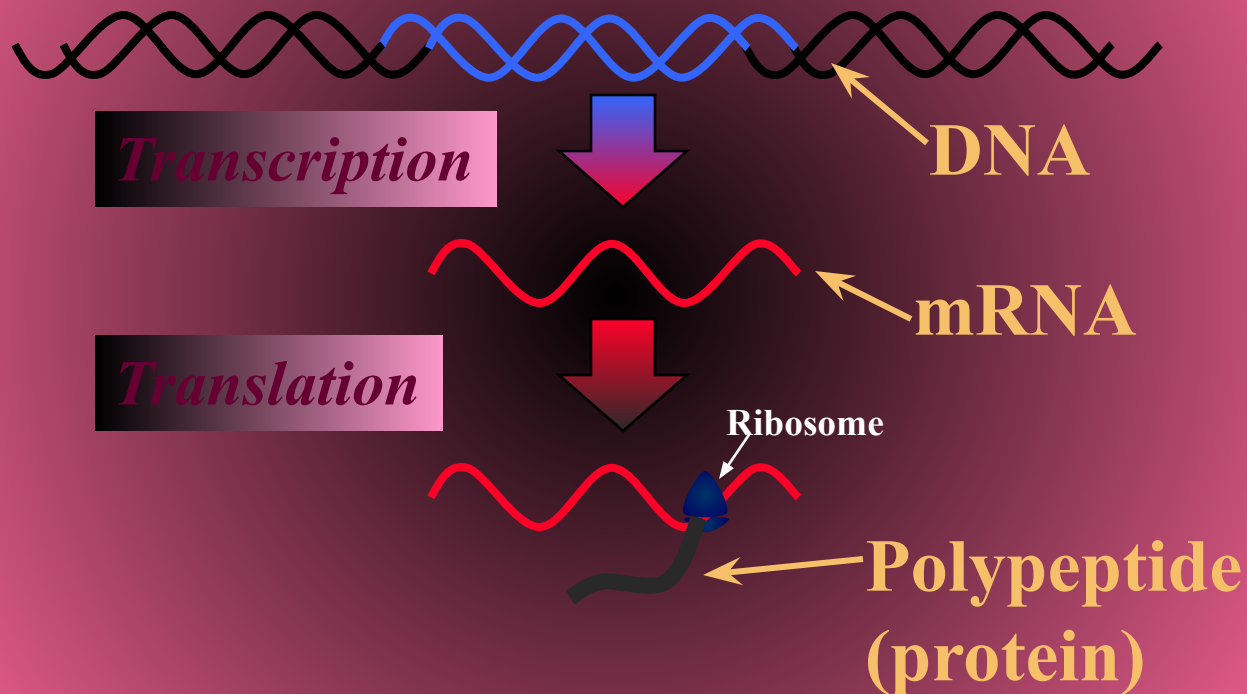


Introduction

The Central Dogma of Molecular Biology

Cell



Protein Synthesis

Flow of Information:



Transcription is the process by which a molecule of DNA is copied into a complementary strand of RNA.

This is called messenger RNA (mRNA) because it acts as a messenger between DNA and the ribosomes where protein synthesis is carried out.

Protein Synthesis Transcription

Transcription process

- **RNA polymerase** (an enzyme) attaches to DNA at a special sequence that serves as a “start signal”.
- The DNA strands are separated and one strand serves as a template.
- The **RNA bases** attach to the complementary DNA template, thus synthesizing mRNA.

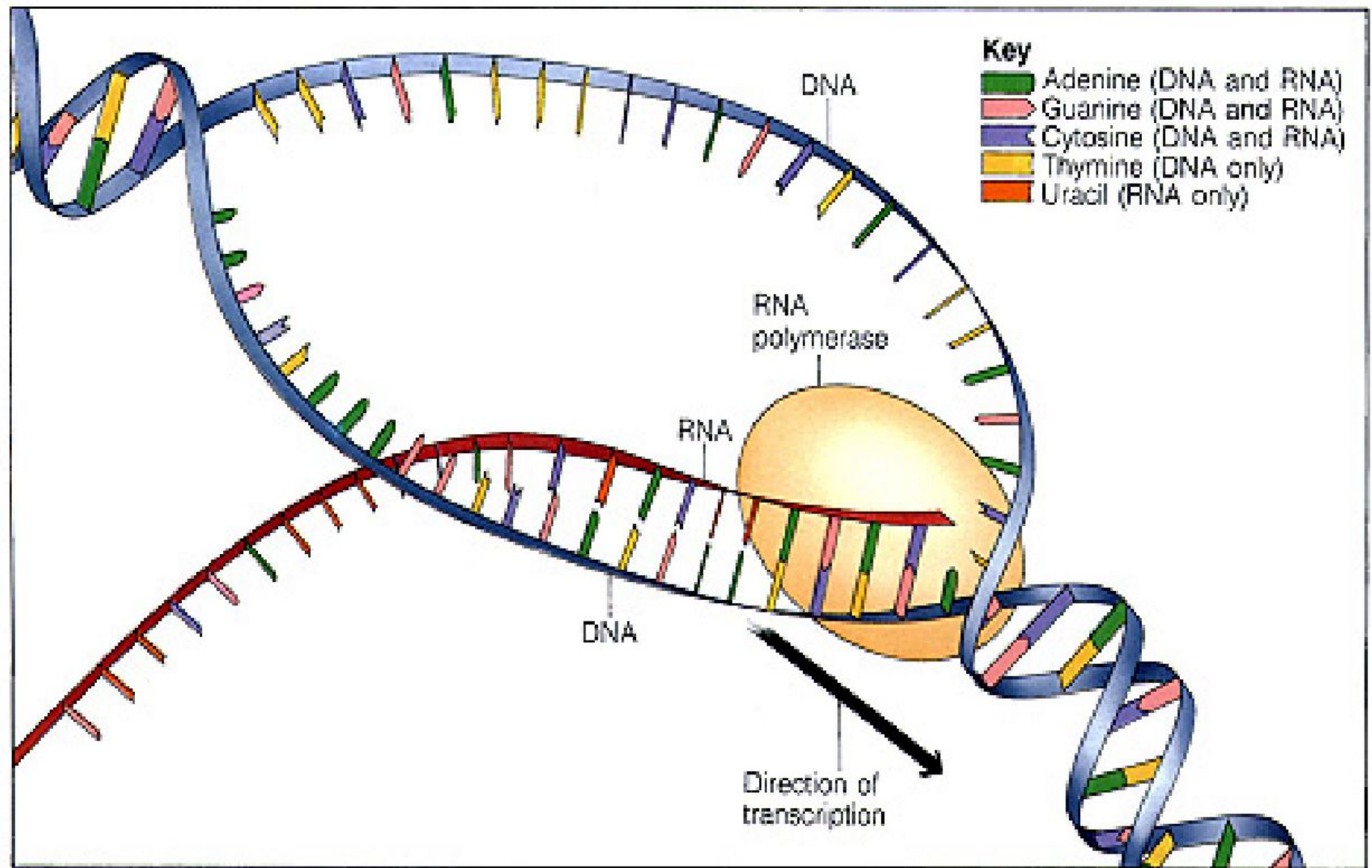
Protein Synthesis: Transcription

Transcription process continued

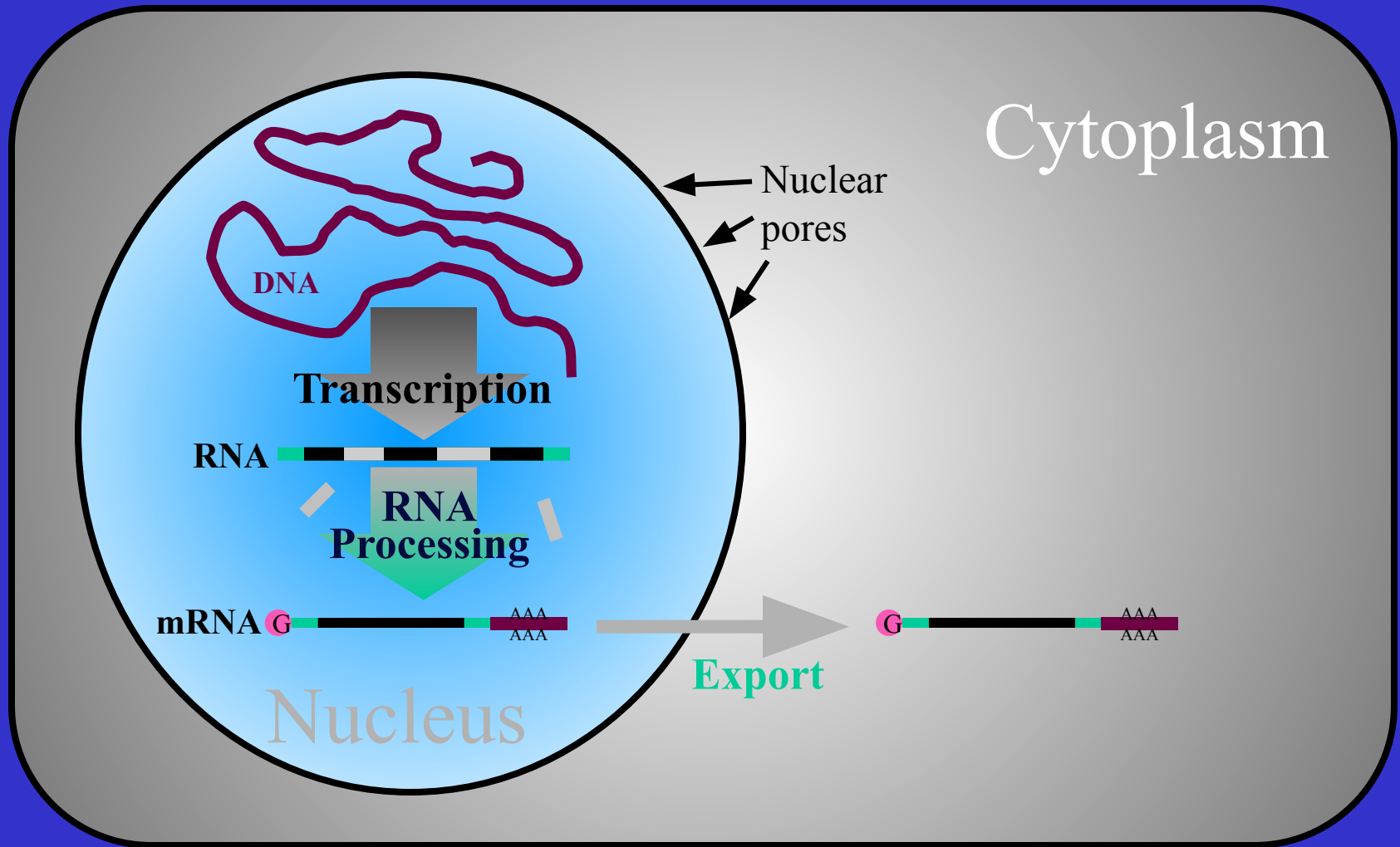
- The RNA polymerase recognizes a termination site on the DNA molecule and releases the new mRNA molecule.

(mRNA leaves the nucleus and travels to the ribosome in the cytoplasm.)

Protein Synthesis: Transcription



Eukaryotic Transcription



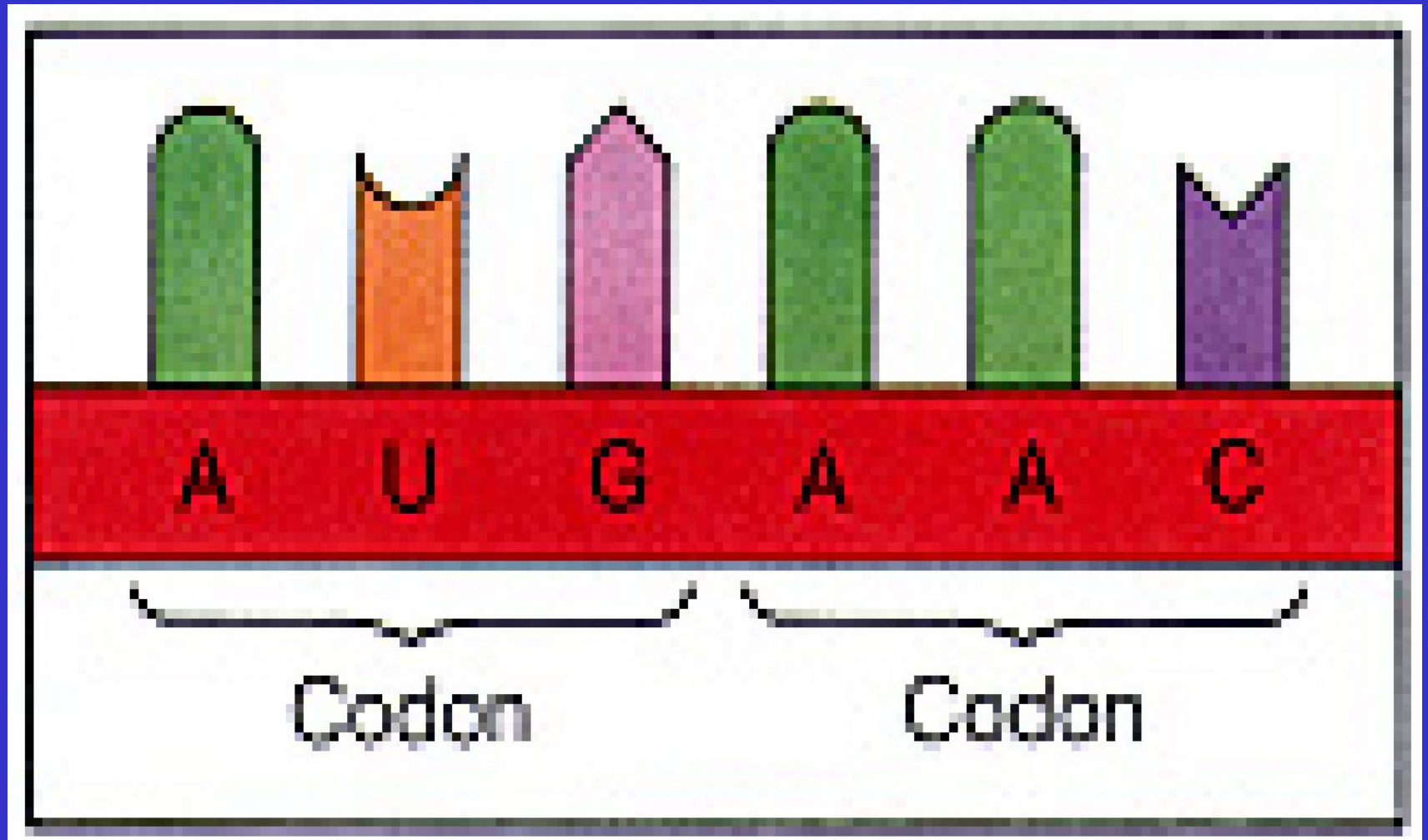
Protein Synthesis: Translation

Translation is the process of decoding a mRNA molecule into a **polypeptide chain or protein.**

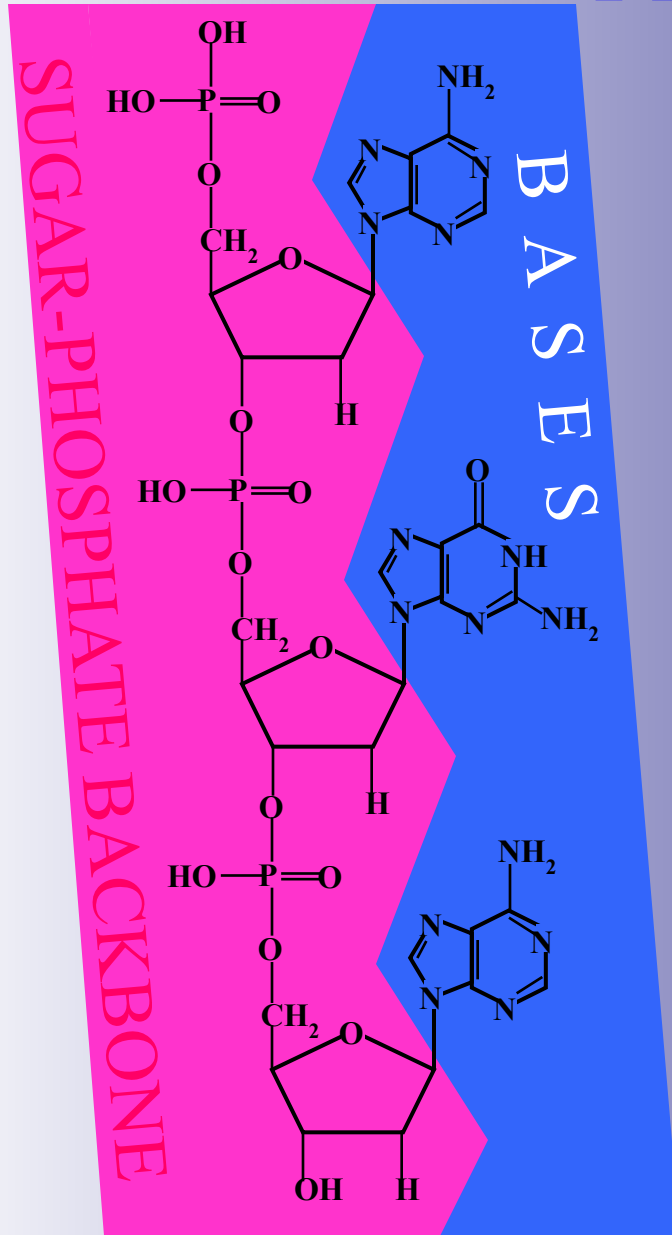
Each combination of 3 nucleotides on mRNA is called a **codon or three-letter code word.**

Each codon specifies a **particular amino acid** that is to be placed in the polypeptide chain (protein).

Protein Synthesis: Translation



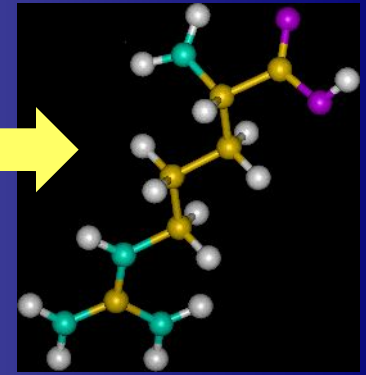
A Codon



Adenine

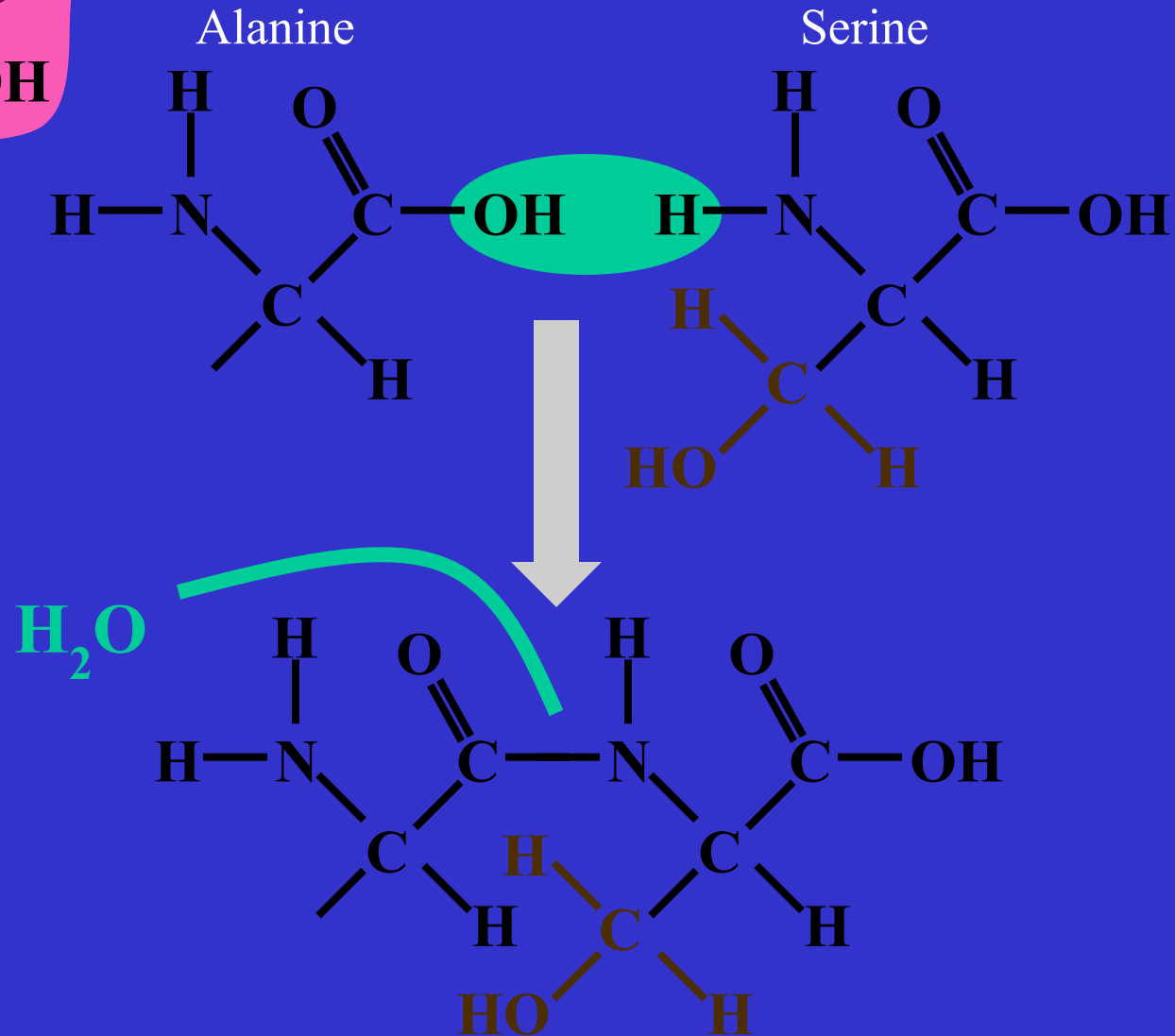
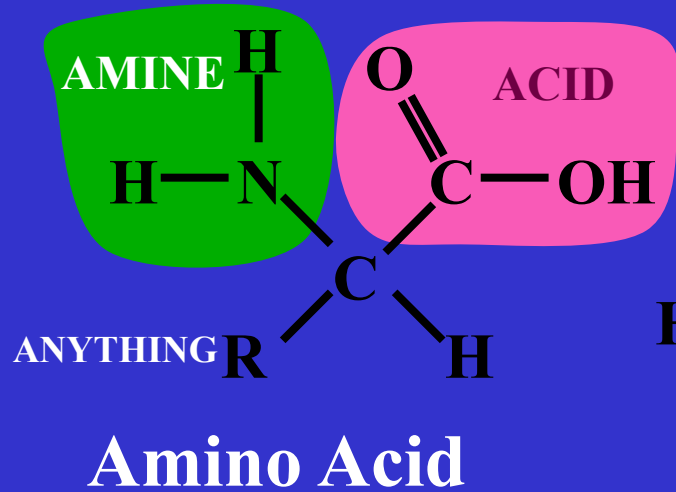
Guanine

Adenine



Arginine

Protein Synthesis



Protein Synthesis: Translation

- A three-letter code is used because there are **20 different amino acids** that are used to make proteins.
- If a **two-letter code** were used there would not be enough codons to select all 20 amino acids.
- That is, there are 4 bases in RNA, so 4^2 (4×4) = 16; **where as 4^3 ($4 \times 4 \times 4$) = 64.**

Protein Synthesis: Translation

Four Code Letters			
A		G	
C		U	
Sixteen Doublets from the Four Code Letters			
AA	AC	AG	AU
CA	CC	CG	CU
GA	GC	GG	GU
UA	UC	UG	UU

Protein Synthesis: Translation

- Therefore, there is a total of 64 codons with mRNA, 61 specify a particular amino acid.
- This means there are more than one codon for each of the 20 amino acids.
- The remaining three codons (UAA, UAG, & UGA) are stop codons, which signify the end of a polypeptide chain (protein).
- Besides selecting the amino acid methionine, the codon AUG also serves as the “initiator” codon, which starts the synthesis of a protein

Protein Synthesis: Translation

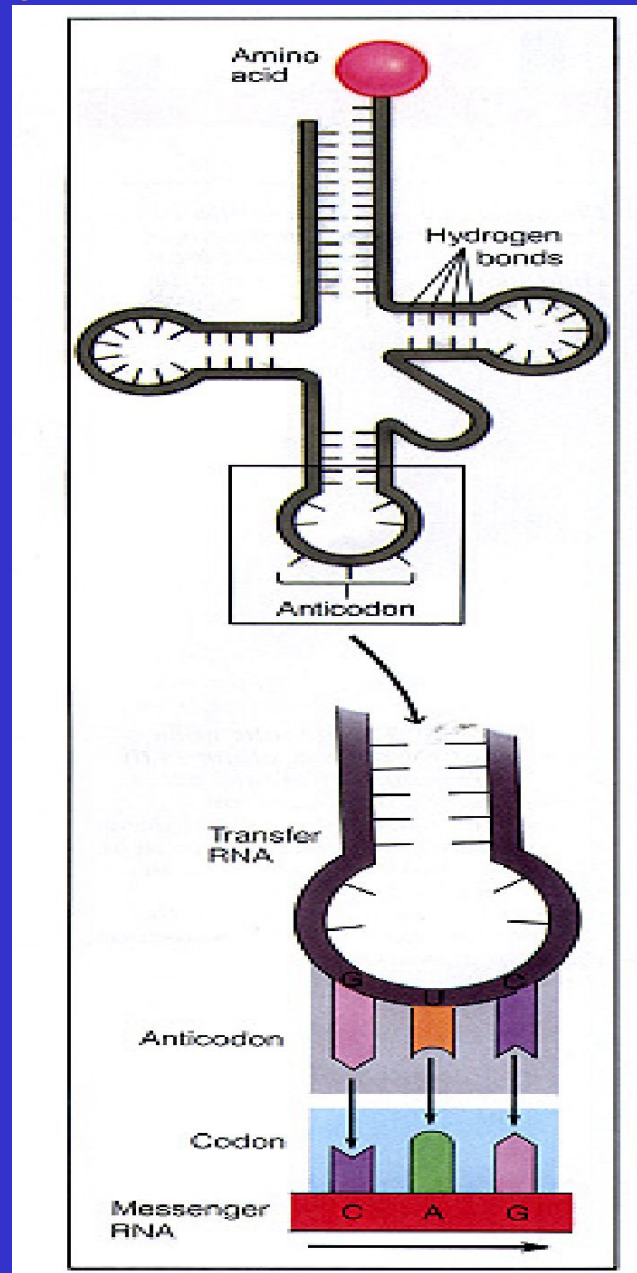
First Base in Code Word	A	Lysine	Arginine	Isoleucine	Threonine	A G U C	Third Base in Code Word
		Lysine	Arginine	Methionine	Threonine		
		Asparagine	Serine	Isoleucine	Threonine		
		Asparagine	Serine	Isoleucine	Threonine		
	G	Glutamic acid	Glycine	Valine	Alanine	A G U C	
Glutamic acid		Glycine	Valine	Alanine			
Aspartic acid		Glycine	Valine	Alanine			
Aspartic acid		Glycine	Valine	Alanine			
U	"Stop" codon	"Stop" codon	Leucine	Serine	A G U C		
	"Stop" codon	Tryptophan	Leucine	Serine			
	Tyrosine	Cysteine	Phenylalanine	Serine			
	Tyrosine	Cysteine	Phenylalanine	Serine			
C	Glutamine	Arginine	Leucine	Proline	A G U C		
	Glutamine	Arginine	Leucine	Proline			
	Histidine	Arginine	Leucine	Proline			
	Histidine	Arginine	Leucine	Proline			
		A	G	U	C		
Second Base in Code Word							

Protein Synthesis: Translation

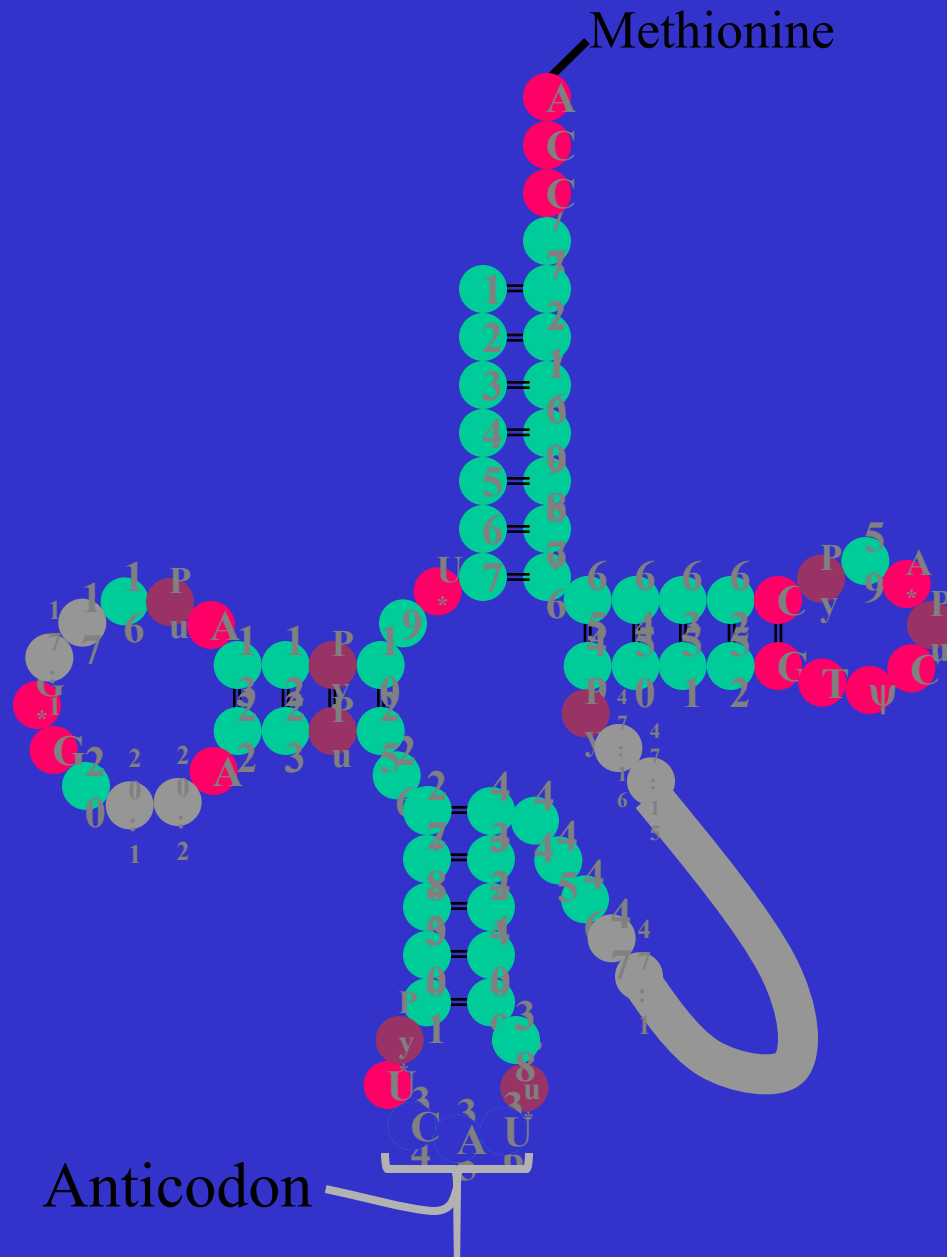
Transfer RNA (tRNA)

- Each tRNA molecule has 2 important sites of attachment.
- One site, called the anticodon, binds to the codon on the mRNA molecule.
- The other site attaches to a particular amino acid.
- During protein synthesis, the anticodon of a tRNA molecule base pairs with the appropriate mRNA codon.

Protein Synthesis: Translation



Met-tRNA



Protein Synthesis: Translation

Ribosome:

- Are made up of 2 subunits, a large one and a smaller one, each subunit contains ribosomal RNA (rRNA) & proteins.
- Protein synthesis starts when the two subunits bind to mRNA.
- The initiator codon AUG binds to the first anticodon of tRNA, signaling the start of a protein.

Protein Synthesis: Translation

Ribosome:

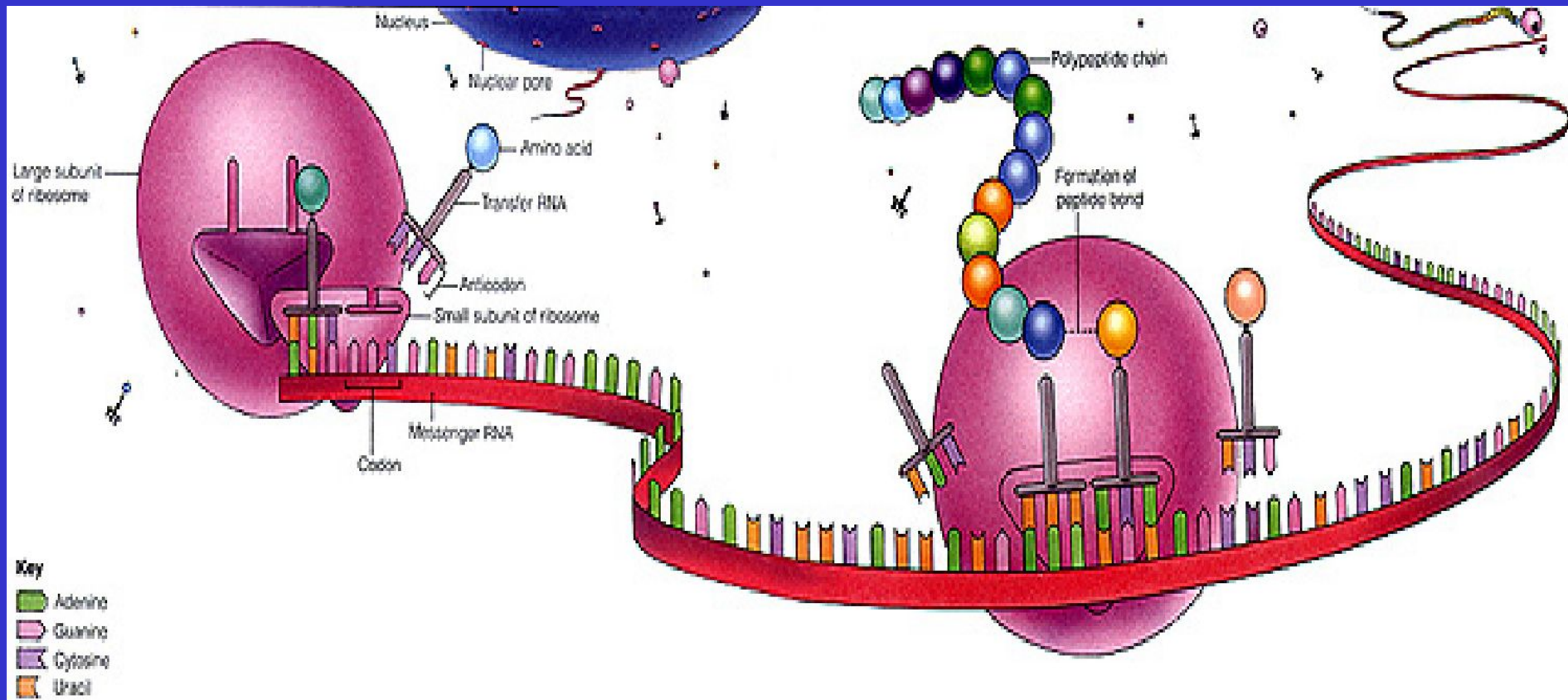
- The anticodon of another tRNA binds to the next mRNA codon, bringing the **2nd amino acid** to be placed in the protein.
- As each anticodon & codon bind together a **peptide bond forms between the two amino acids.**

Protein Synthesis: Translation

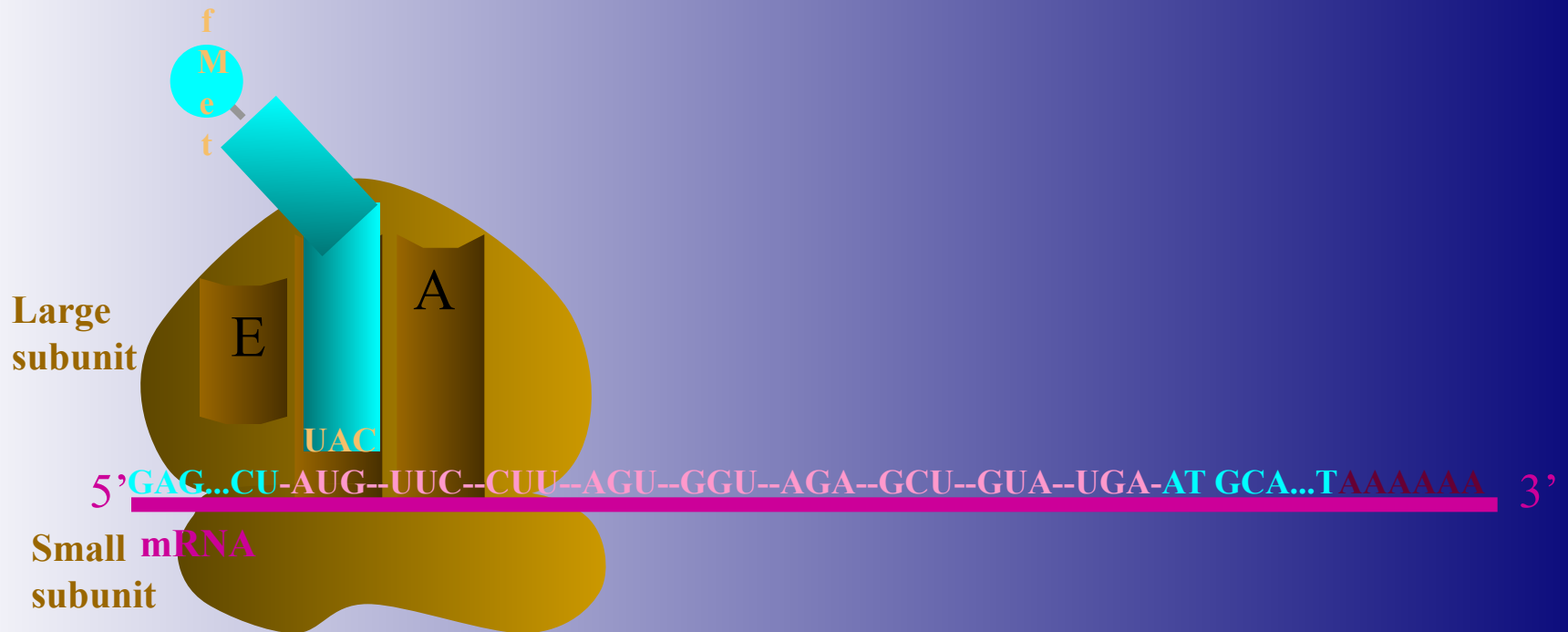
Ribosome:

- The protein chain continues to grow until ribosome reaches the stop codon, which results in the release of the new protein and mRNA, completing the process of translation.

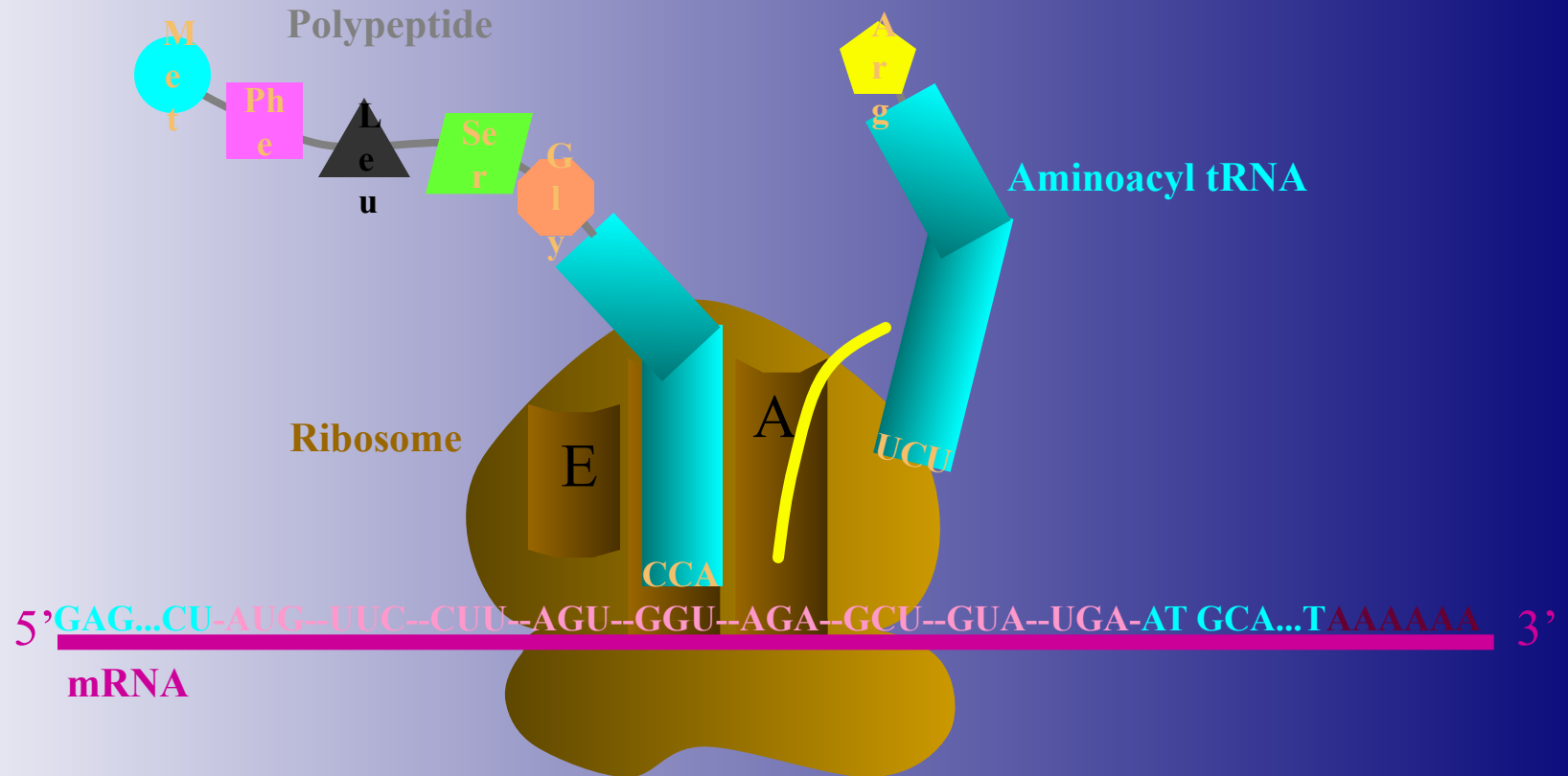
Protein Synthesis: Translation



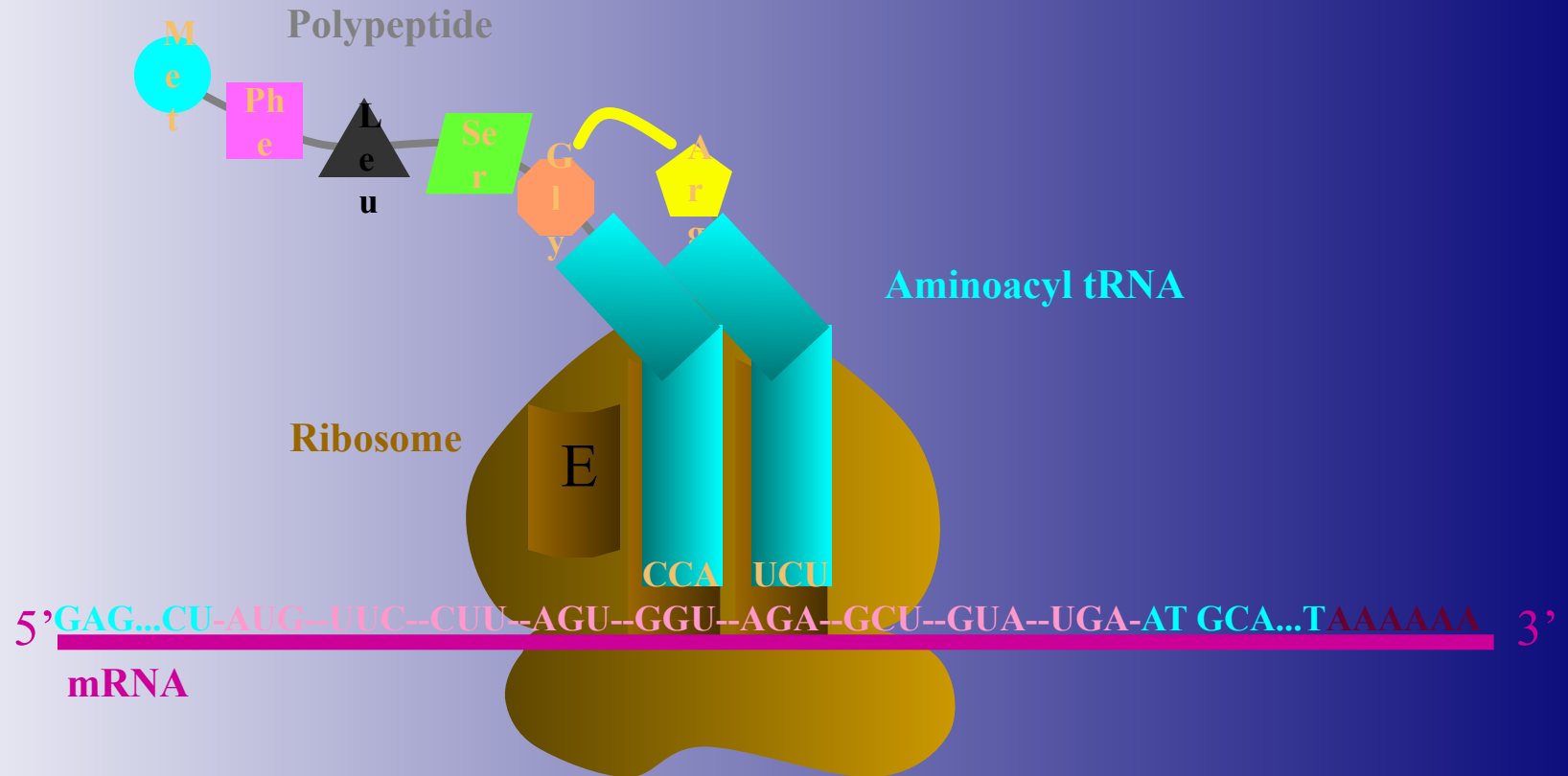
Translation - Initiation



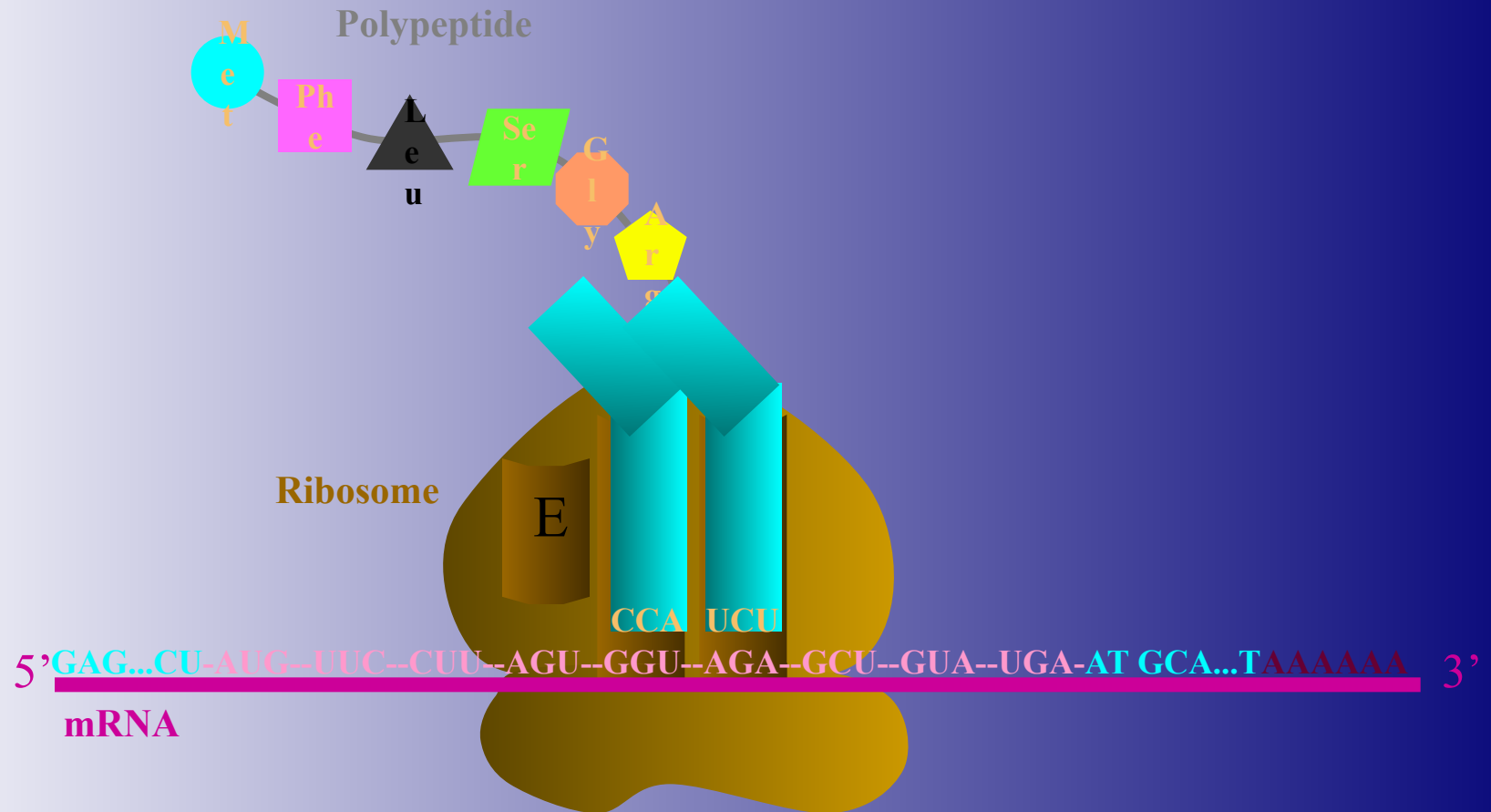
Translation - Elongation



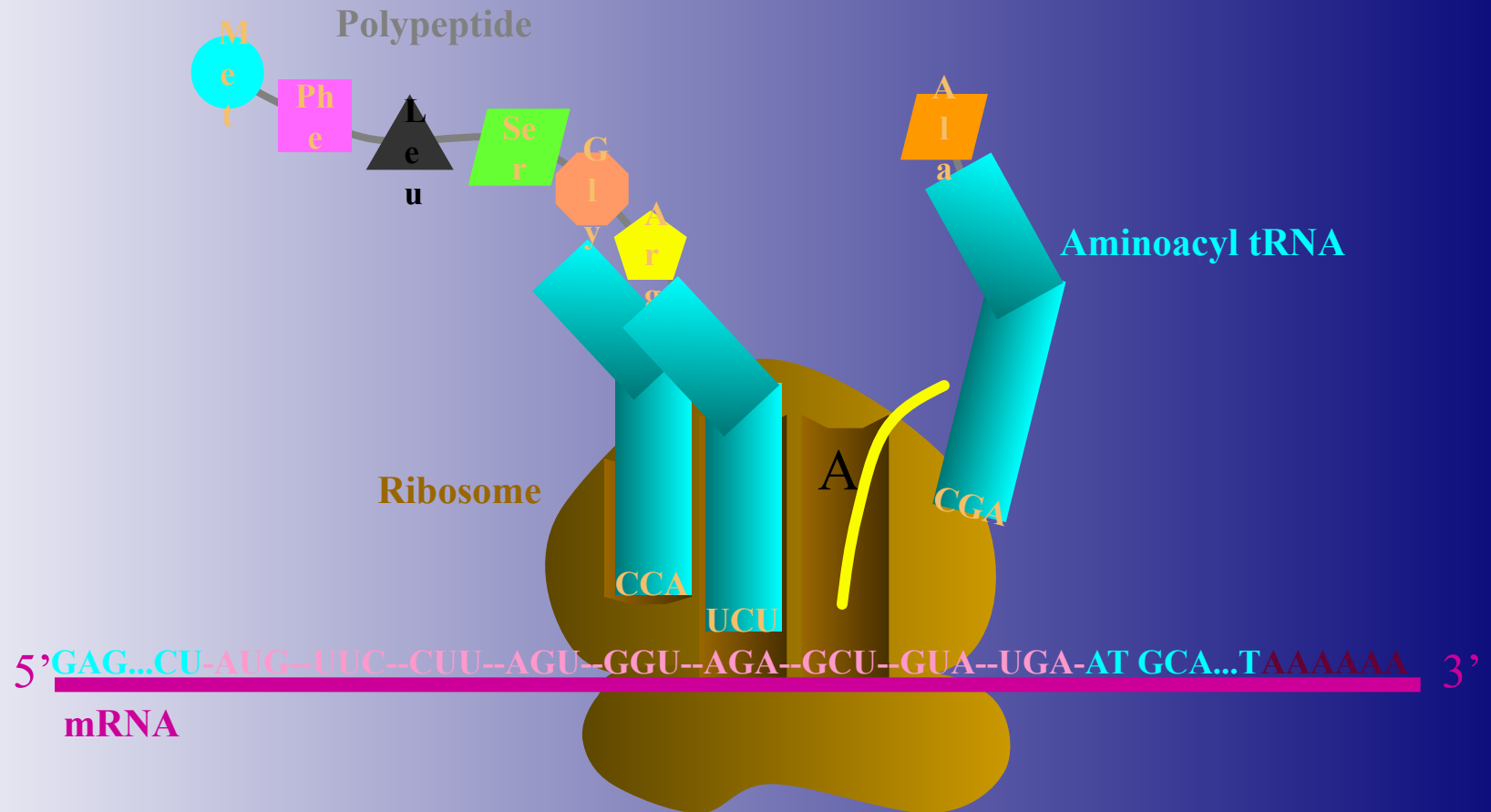
Translation - Elongation



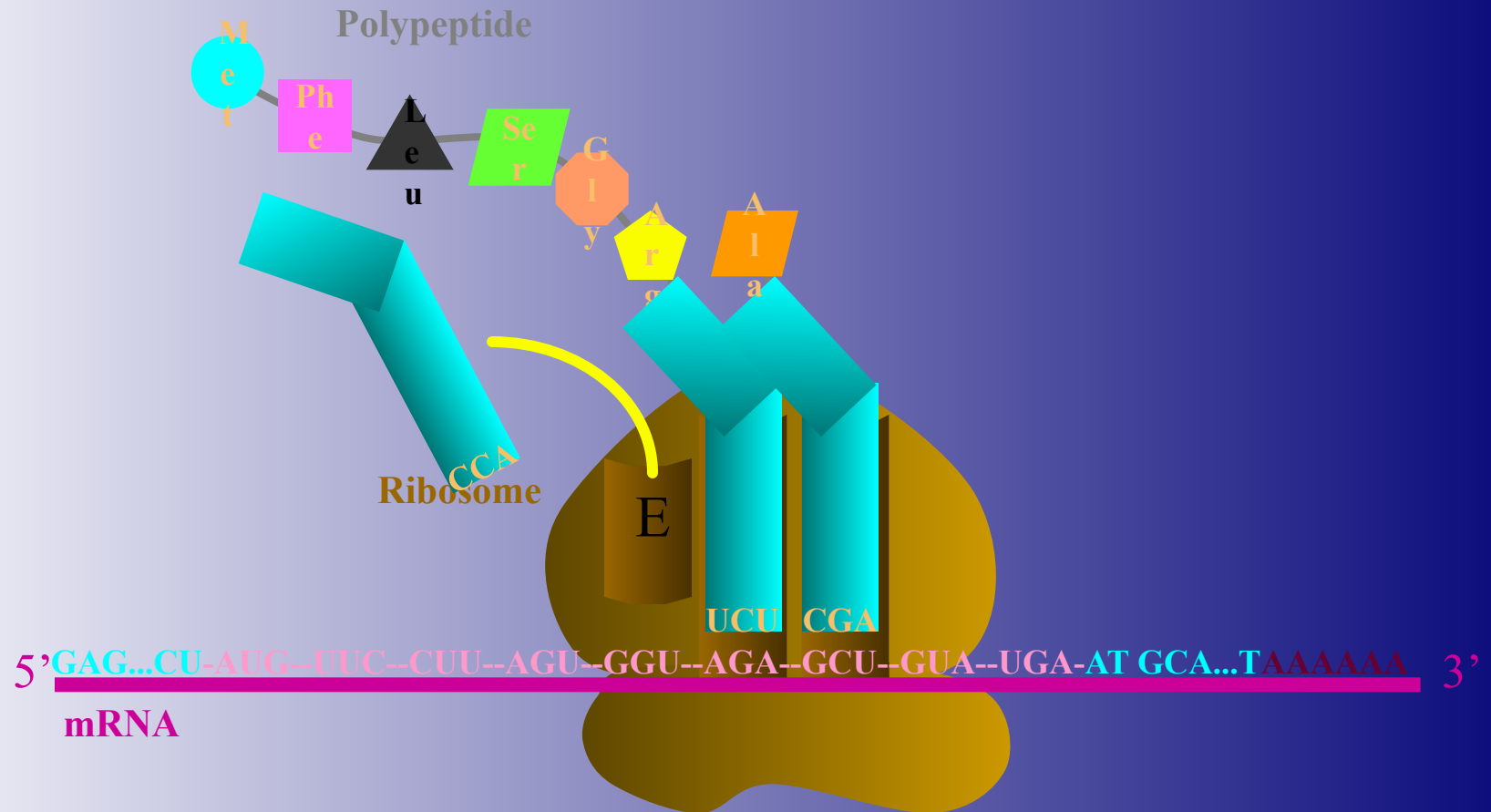
Translation - Elongation



Translation - Elongation



Translation - Elongation



Transcription And Translation In Prokaryotes

